

1.5 Ratio and proportion

Subject content	Guidance	Curriculum reference
A Write and interpret ratios written in notation	A class contains 15 boys and 12 girls. What is the ratio of boys to girls?	N7.5B
B Simplify and use ratios, including integers, decimals and fractions	Write 13:65 in its simplest form	N7.5C N8.5A
C Write and compare ratios in unitary form ($1:n$ or $n:1$)	Write 16:30 in the form $1:n$	N8.5C
D Divide a quantity into two or three parts	Divide \$160 in the ratio 4:3:1 Divide \$160 in the ratio 4:1	N7.5D N8.5B
E Solve word problems that involve dividing a quantity into two or three parts	A man wins \$1 000 in a competition. He shares his winnings between his 3 children, Ana, Boris and Carl, in the ratio 5:6:9 How much more than Ana does Carl get?	N7.5D N8.5D N9.4C

1.5 Ratio and proportion continued

Subject content	Guidance	Curriculum reference
F Solve word problems using ratios to find missing quantities	The ratio of boys to girls in a school is 8:7 There are 176 boys in the school. How many girls are there in the school?	N7.5E N8.5D N9.4C
G Use fractions to describe and compare proportions	A student scores 39 out of 50 on a German test. The same student gets 16 questions out of 20 correct on an English test. Which test did the student do better in?	N7.5F
H Use percentages to describe and compare proportions	A student scores 58% on a German test. The same student gets 21 questions out of 37 correct on an English test. Which test did the student do better in?	N7.5G
I Understand and use the relationship between ratio and proportion	The ratio of men to women in a gym is 9:4 What fraction of the people in the gym are men? What percentage of the people in the gym are women?	N7.5H
J Solve simple problems involving direct proportion in context	8 pencils cost \$1.36 How much would 13 pencils cost?	N7.5A
K Calculate values where quantities vary in direct proportion	When $x=16$, $y=4$ y is directly proportional to x Find the value of x when $y=5$	N9.4A
L Solve word problems using ratio and/or proportion	One man is 1.8 m tall and his foot is 28 cm long. Another man is 1.65 m tall and his foot is 25 cm long. Is foot length of men in direct proportion to their height?	N7.5D N8.5D N9.4C